git + GitHub

AGENDA?

What is Git and GitHub?
Why use git?
Installation.
How to use git.

WHAT IS GIT?

- Git is a 'distributed version control system' - eh?

- That basically means...

  - It's a system that records changes to our files over time

  - We can recally specific versions of those files at any given time

  - Many people can easily collaborate on a project and have their own version of project files on their computer

Slide 2 | Presenter view | Exit

WHY USE GIT?

Project X for client

WHY USE GIT?

Project X for client

Slide 4 | Presenter view | Exit

WHY USE GIT?

Project X for client

Slide 5 | Presenter view | Exit

WHY USE GIT?

Project X for client

Slide 6 | Presenter view | Exit

WHY USE GIT?

Project X for client

Slide 7 | Presenter view | Exit

WHY USE GIT?

Project X for client - v1

Project X for client - v2

Project X for client - v3

Project X for client - v4

Slide 8 | Presenter view | Exit

WHY USE GIT?

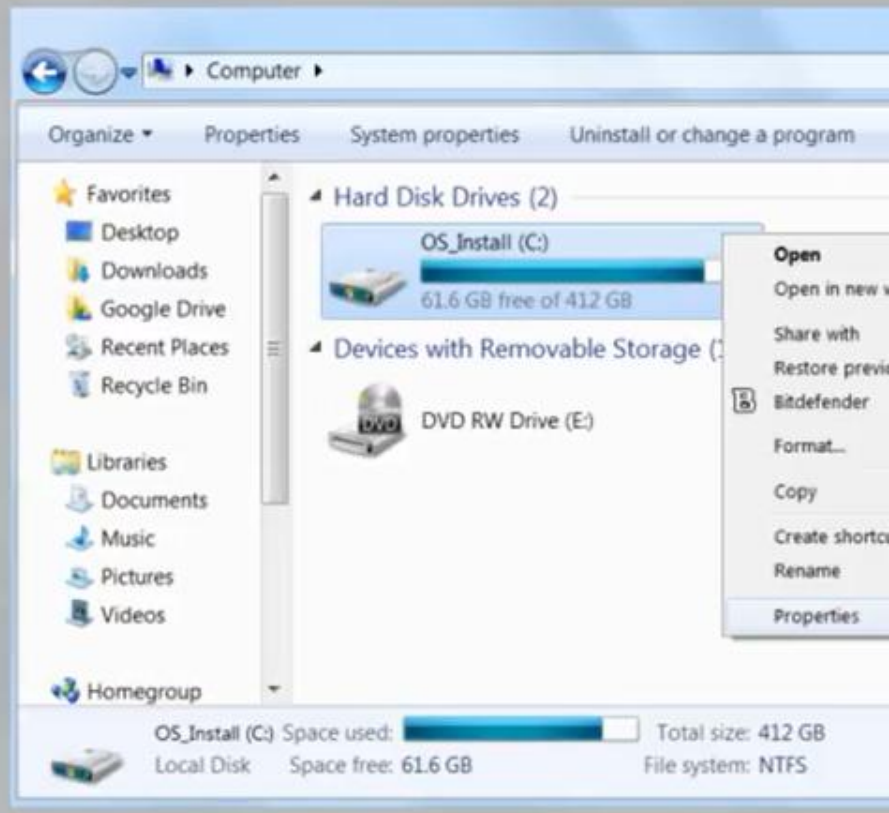
Project X for client - v1

Project X for client - v2

Project X for client - v3

...several days / emails later....

Project X for client - v25 !!!



WHY USE GIT?

- Store revisions in a project history in just one directory
- Rewind to any revision in the project I wanted to
- Work on new features without messing up the main codebase
- Easily collaborate with other programmers

Slide 11 | Presenter view | Exit

GITHUB

- Online service that hosts our projects

- Share our code with other developers

- Developers can download the projects and work on them

- They can re-upload their edits and merge them with the main codebase

Slide 12 | Presenter view | Exit

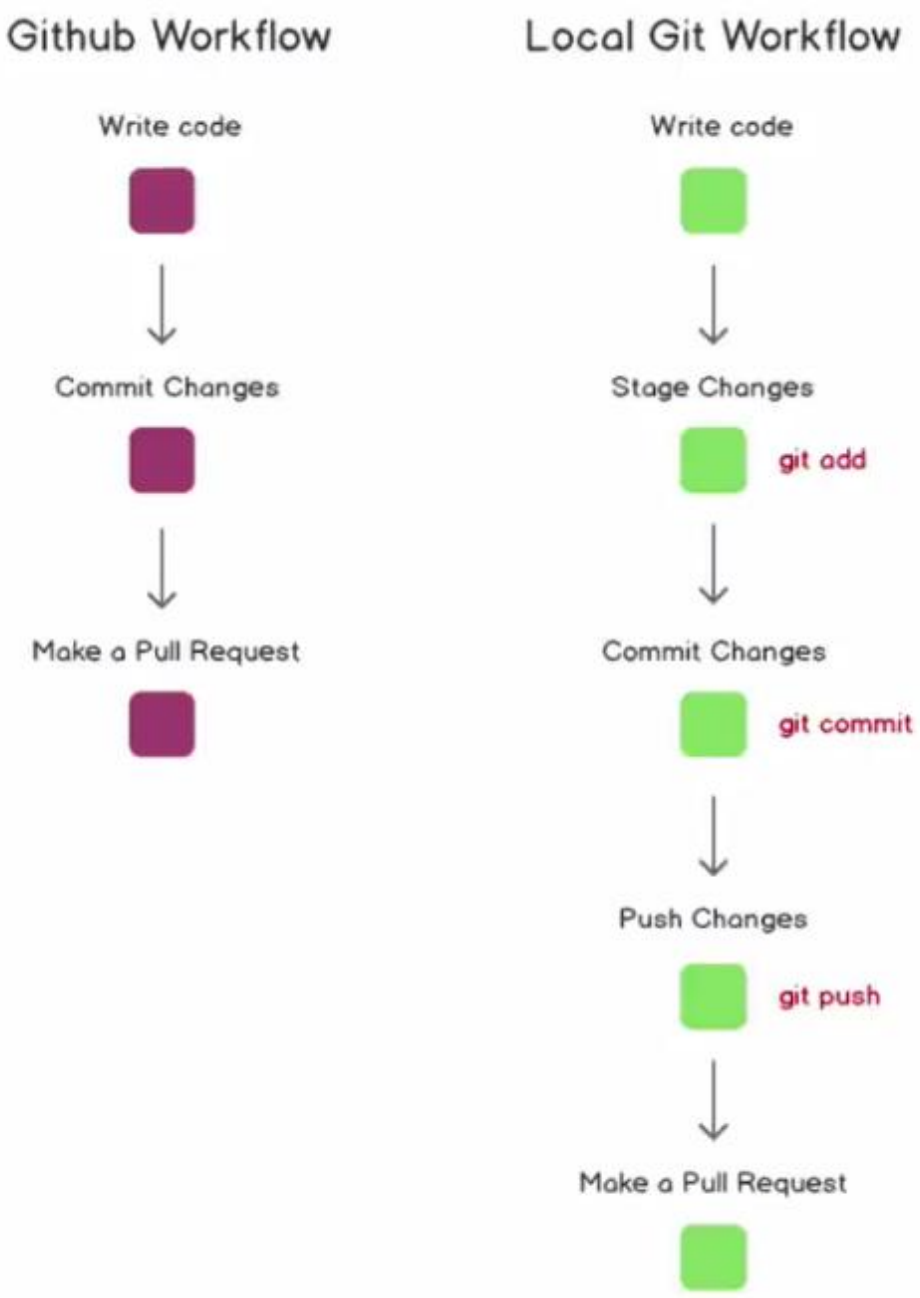
Terms

- Directory -> Folder
- Terminal or Command Line -> Interface for Text Commands
- CLI -> Command Line Interface
- cd -> Change Directory
- Code Editor -> Word Processor for Writing Code
- Repository -> Project, or the folder/place where your project is kept
- Github -> A website to host your repositories online

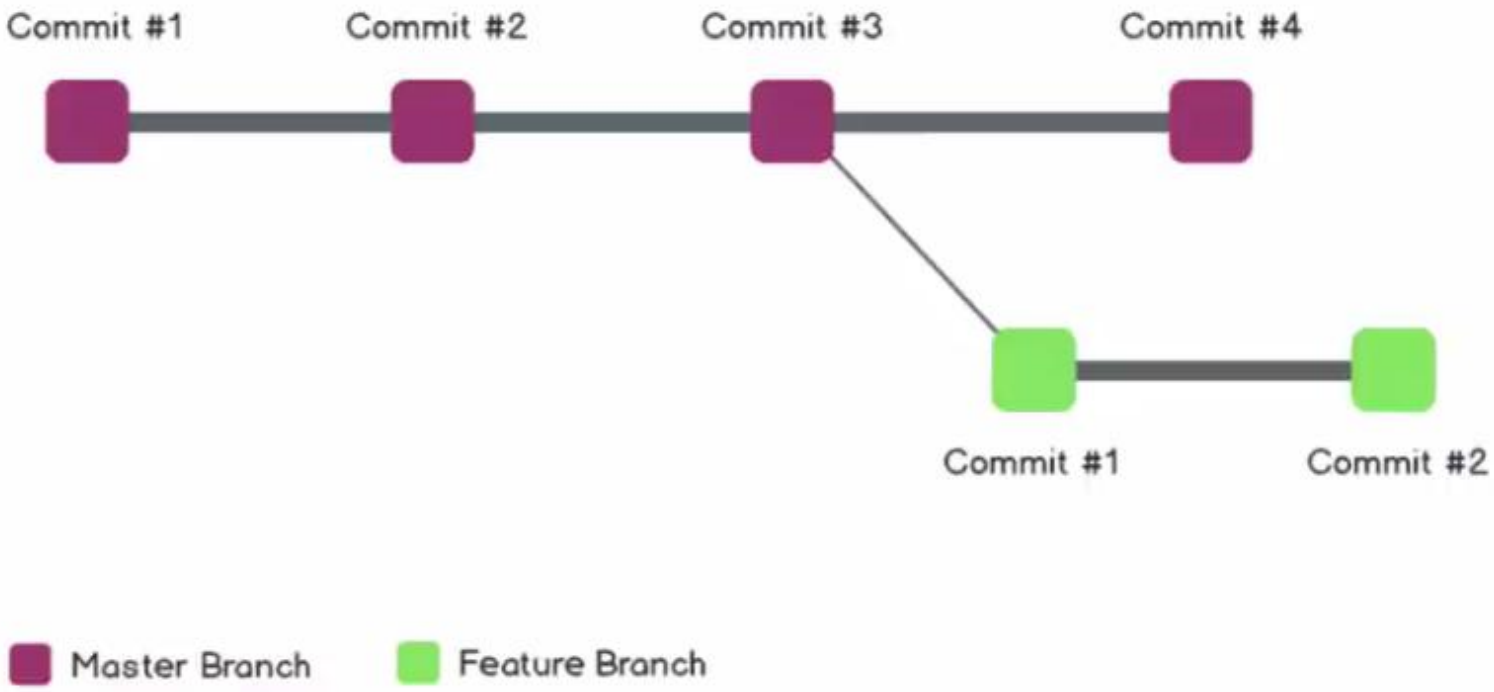
Git Commands

- Clone -> Bring a repository that is hosted somewhere like Github into a folder on your local machine
- add -> Track your files and changes in Git
- commit -> Save your files in Git
- push -> Upload Git commits to a remote repo, like Github
- pull -> Download changes from remote repo to your local machine, the opposite of push

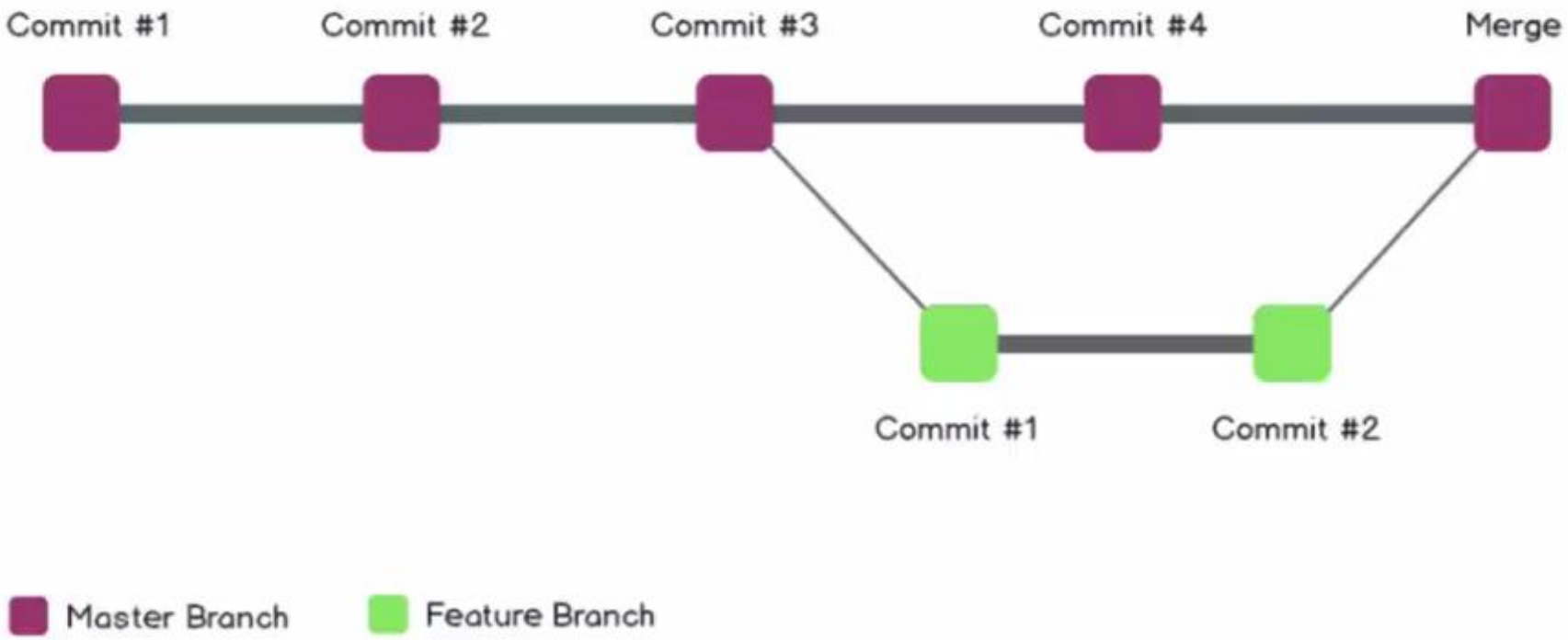
- repo -> repository
- `clone` -> bring a repo down from the internet (remote repository like Github) to your local machine
- `add` -> track your files and changes with Git
- `commit` -> save your changes into Git
- `push` -> push your changes to your remote repo on Github (or another website)
- `pull` -> pull changes down from the remote repo to your local machine
- `status` -> check to see which files are being tracked or need to be commited
- `init` -> use this command inside of your project to turn it into a Git repository and start using Git with that codebase


Github Workflow
Write code
Commit Changes
Make a Pull Request
Local Git Workflow
Write code
Stage Changes
git add
Commit Changes
git commit
Push Changes
git push
Make a Pull Request

Git Branching



Git Branching



git --distributed-even-if-your-workflow-isnt

Search entire site...

About

Documentation

Blog

Downloads

GUI Clients

Logos

Community

The entire **Pro Git book** written by Scott Chacon and Ben Straub is available to read online for free. Dead tree versions are available on Amazon.com.

Downloads

Mac OS X Windows

Linux Solaris

Older releases are available and the Git source repository is on GitHub.

GUI Clients

Git comes with built-in GUI tools (**git-gui**, **gitk**), but there are several third-party tools for users looking for a platform-specific experience.

View GUI Clients →

Logos

Various Git logos in PNG (bitmap) and EPS (vector) formats are available for use in online and print projects.

View Logos →

Git via Git

ben — -zsh — 80×24

```
Last login: Fri Apr 10 13:15:05 on ttys001
ben@Benjamin ~ % git --version
git version 2.39.5 (Apple Git-154)
ben@Benjamin ~ %
```

ben — -zsh — 80×24

```
Last login: Fri Apr 10 13:15:05 on ttys001
ben@Benjamin ~ % git --version
git version 2.39.5 (Apple Git-154)
ben@Benjamin ~ % git help
usage: git [-v | --version] [-h | --help] [-C <path>] [-c <name>=<value>]
           [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
           [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--bare]
           [--git-dir=<path>] [--work-tree=<path>] [--namespace=<name>]
           [--super-prefix=<path>] [--config-env=<name>=<envvar>]
           <command> [<args>]

These are common Git commands used in various situations:

start a working area (see also: git help tutorial)
   clone     Clone a repository into a new directory
   init      Create an empty Git repository or reinitialize an existing one

work on the current change (see also: git help everyday)
   add       Add file contents to the index
   mv        Move or rename a file, a directory, or a symlink
   restore   Restore working tree files
   rm        Remove files from the working tree and from the index

examine the history and state (see also: git help revisions)
```

```
$ git config --global user.email "email@example.com"
```

```
$ git config --global user.name "your name"
```

Advanced

Create a new branch

Let's try something a little more advanced. Say you want to make a new feature but are worried about making changes to the main project while developing the feature. This is where git branches come in.

Branches allow you to move back and forth between 'states' of a project. Official git docs describe branches this way: ‘A branch in Git is simply a lightweight movable pointer to one of

Advanced

Create a new branch

- git checkout -b <branch-name>
- Switch and merge: git checkout master/git switch master (or main, or your primary branch)
- git merge <branch-name>
- Then git push origin master

Optionally, you can delete the branch

- Locally: git branch -d <branch-name>
- Remotely: git push origin --delete <branch-name>

Roll back to a previous state

Temporarily go back (without changing history)

If you just want to check out a previous commit temporarily:

- git checkout <commit_hash> (get hash with git log –oneline)

Create a new commit that reverts a previous one

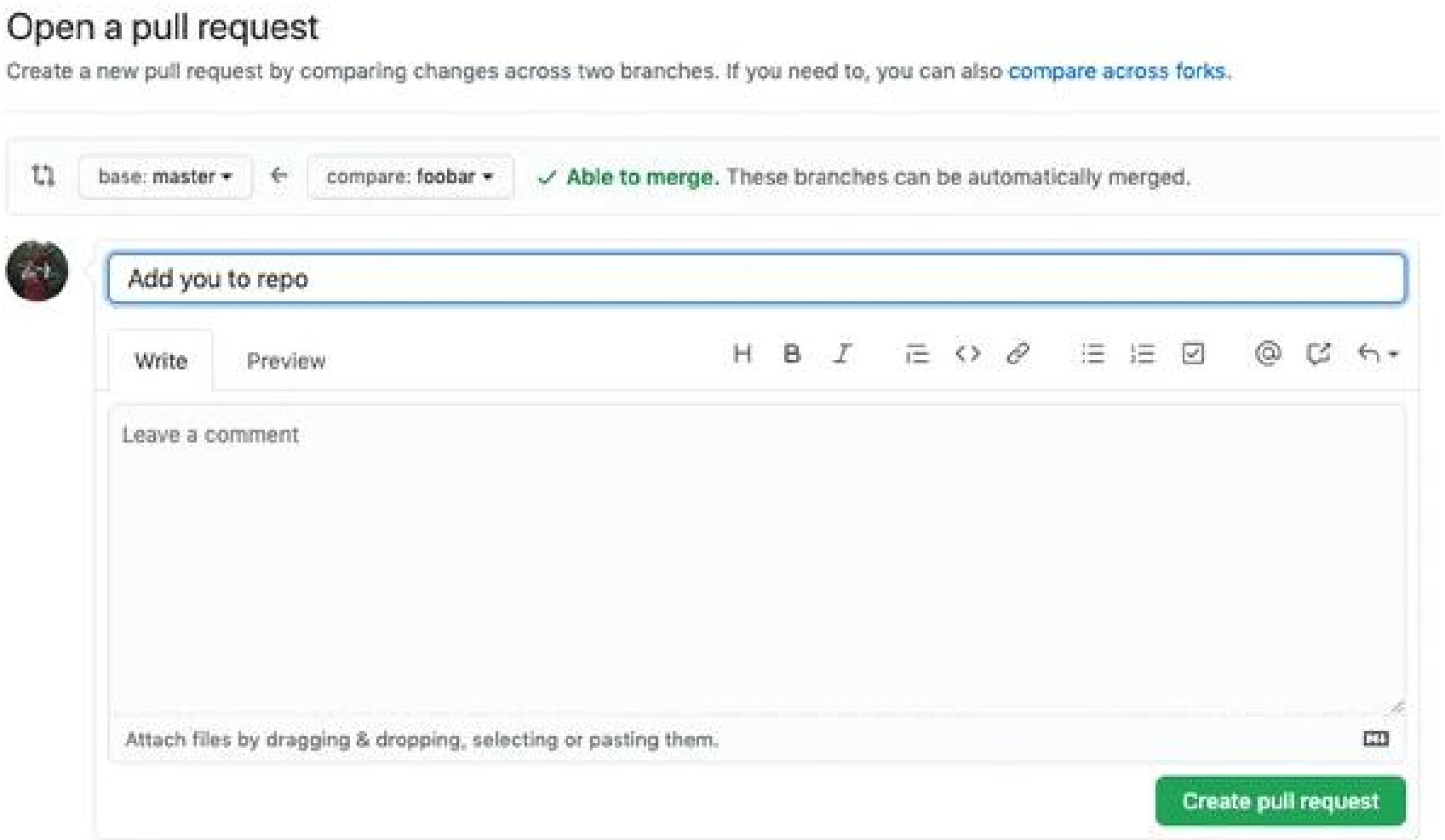
If the commit is already pushed and you want to undo it safely, use:

git revert <1a11e97>. This creates a new commit that reverses the effects of the given commit. Preserves

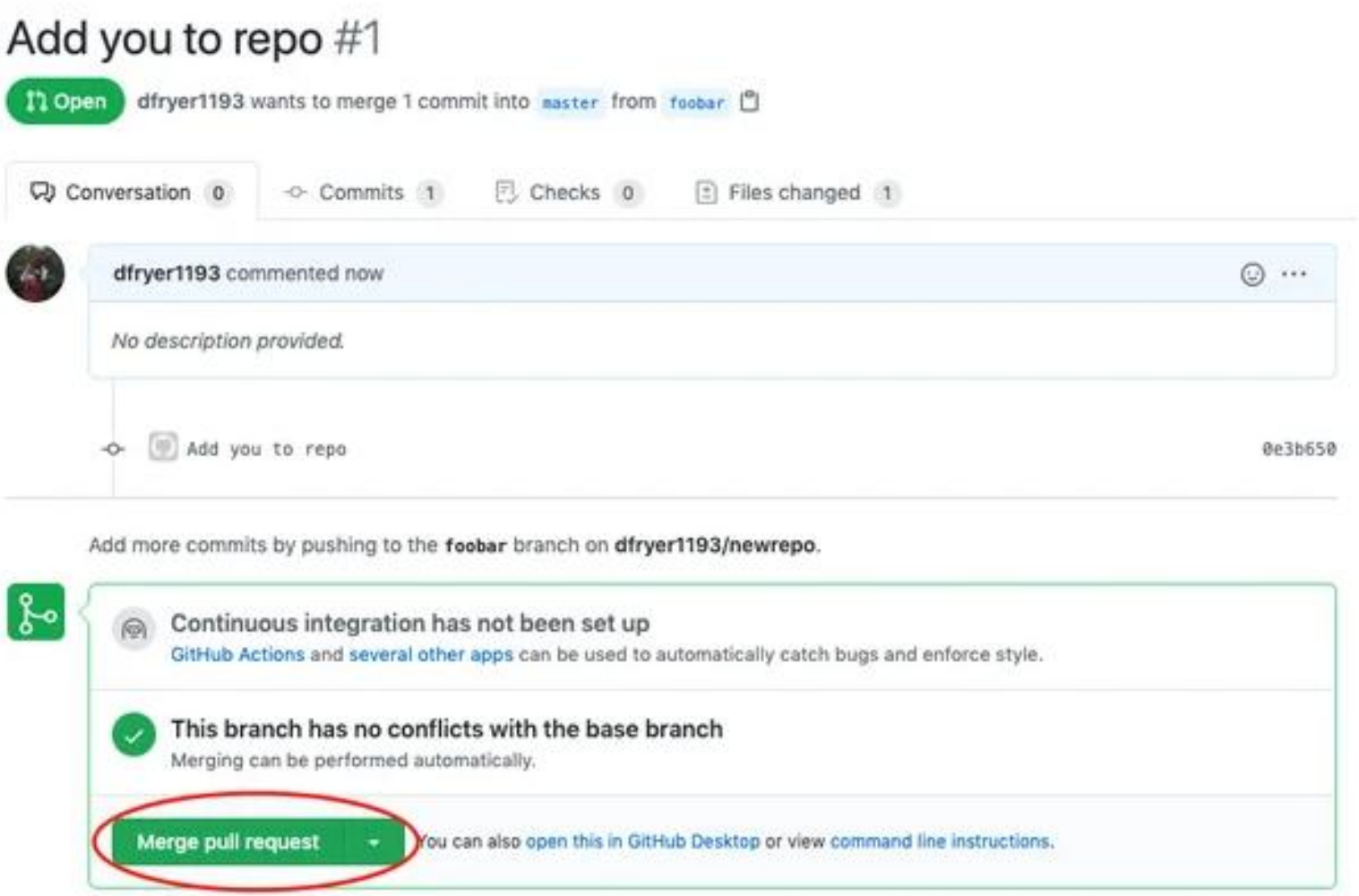
Create a pull request (PR)

A pull request (or PR) is a way to alert a repo's owners that you want to make some changes to their code. It allows them to review the code and make sure it looks good before putting your changes on the primary branch.

This is what the PR page looks like before you've submitted it:



And this is what it looks like once you've submitted the PR request:



Get changes on GitHub Locally

Right now, the repo on GitHub looks a little different than what you have on your local machine. For example, the commit you made in your branch and merged into the primary branch doesn't exist in the primary branch on your local machine.

In order to get the most recent changes that you or others have merged on GitHub, use the git pull origin master command (when working on the primary branch). In most cases, this can be shortened to "**git pull**".

- git pull origin master

QUESTIONS?